

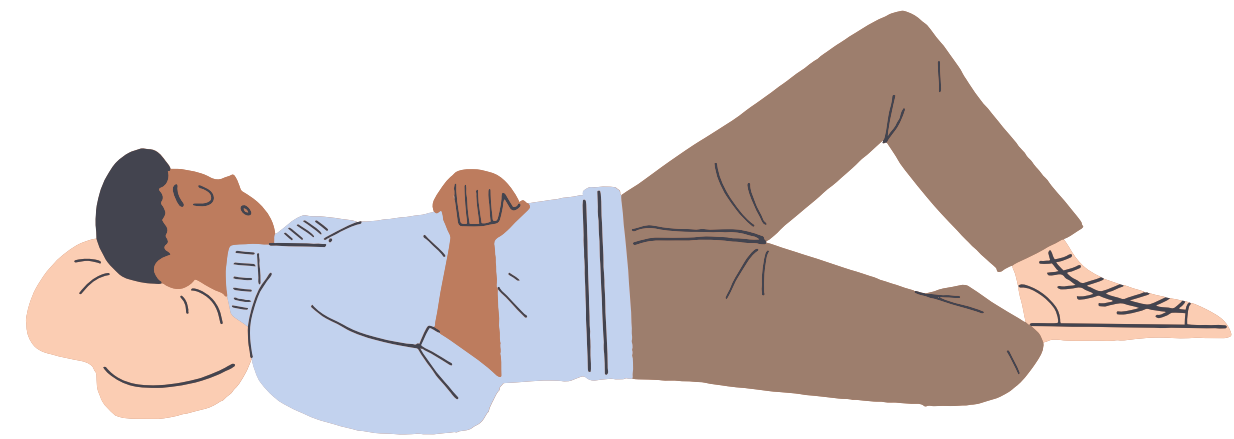


HUMAN ACTIVITY RECOGNITION WITH MACHINE LEARNING ALGORITHMS

Yesim Nur Tortop 911967

AIM OF THE PROJECT

Human Activity Recognition project is to leverage machine learning and sensor technology to automatically recognize, classify, and understand human activities



DATASET DETAILS

21 SUBJECT

6 ACTIVITY

STANDING

WALKING UPSTAIRS

WALKING

WALKING DOWNSTAIRS

SITTING

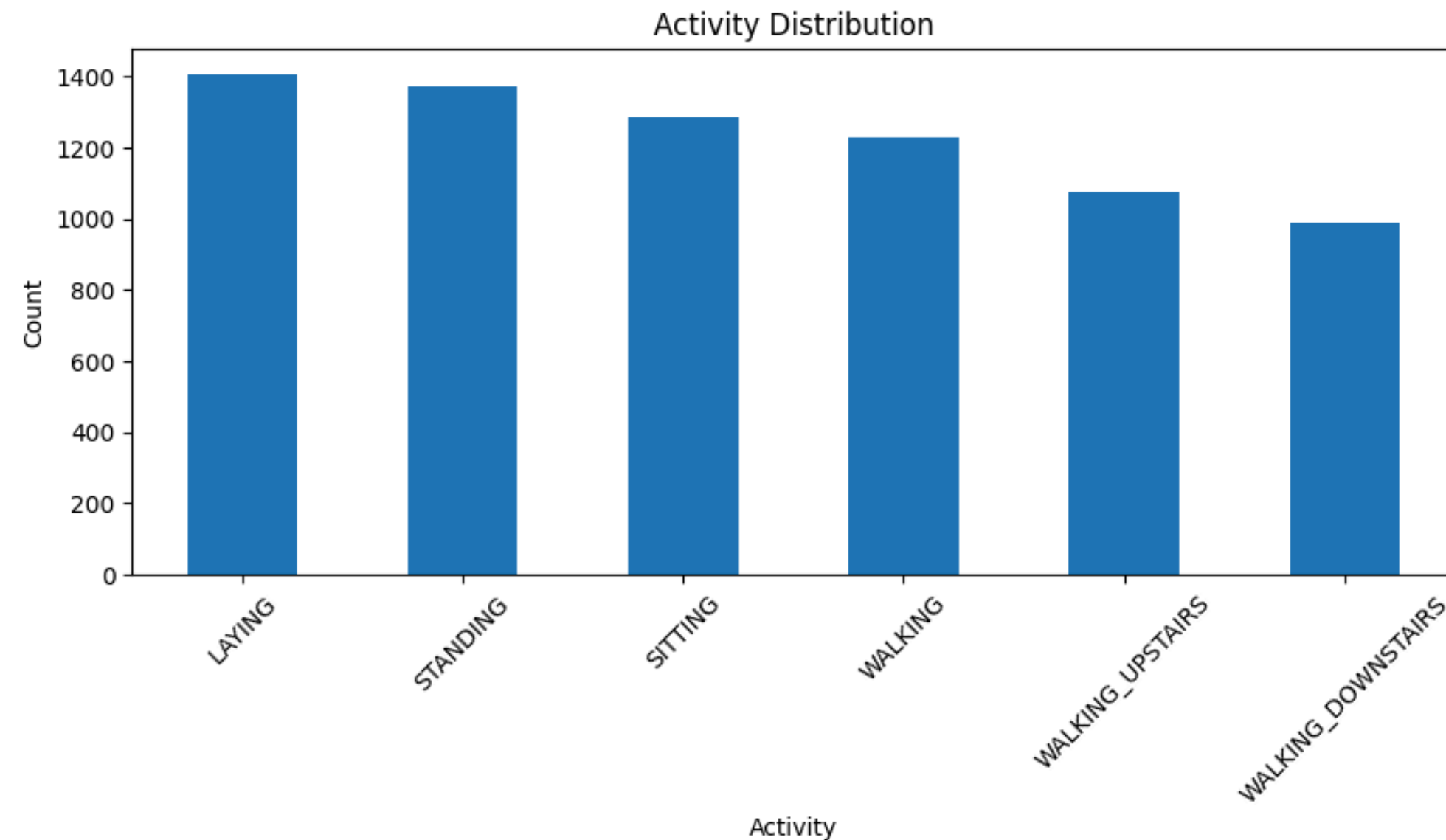
LAYING



DATASET DETAILS

7352 INSTANCE

561 FEATURE



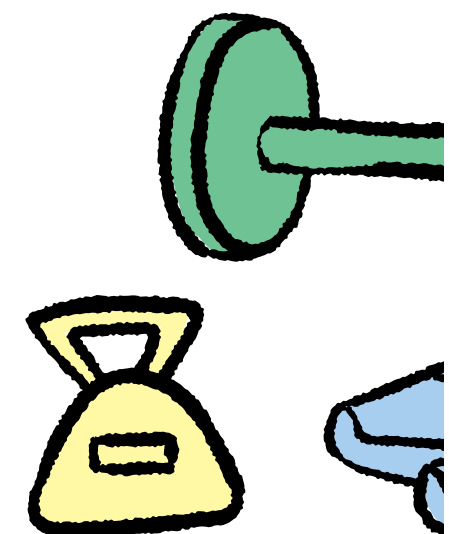
SMOTE

1407 INSTANCE PER
ACTIVITY

DATASET DETAILS

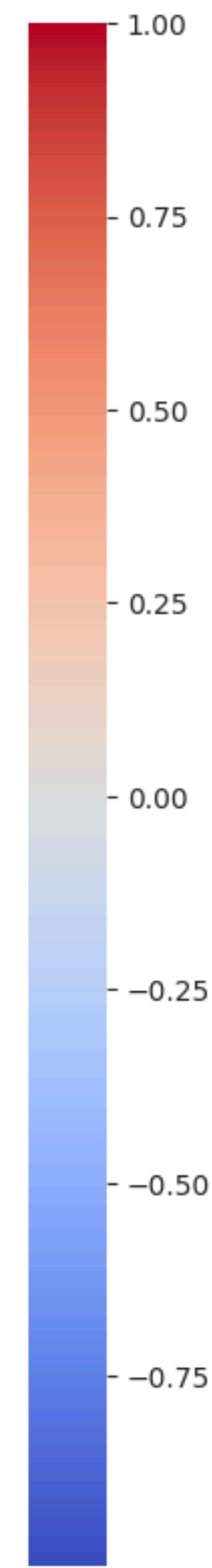
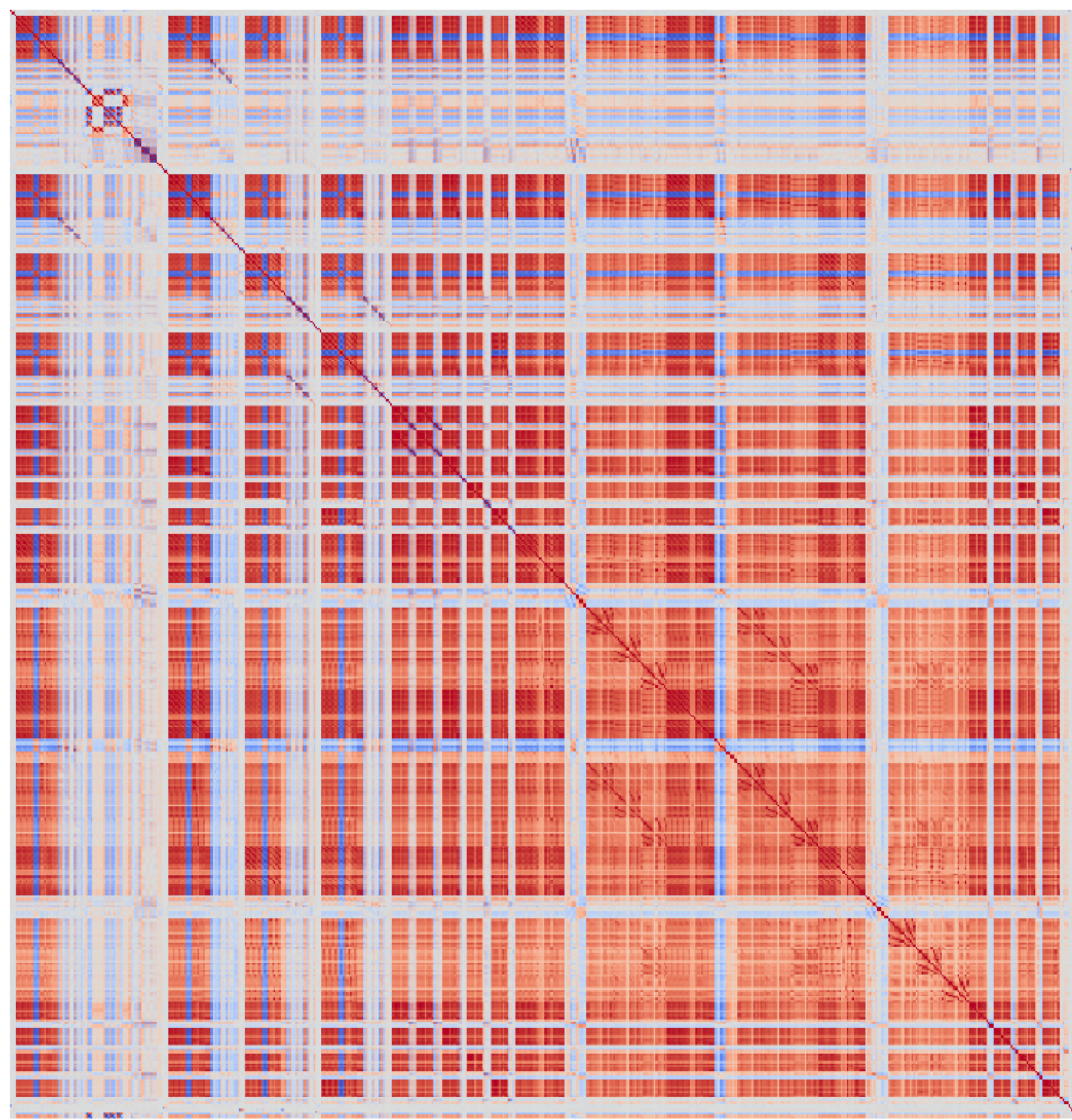
Data provided by each user



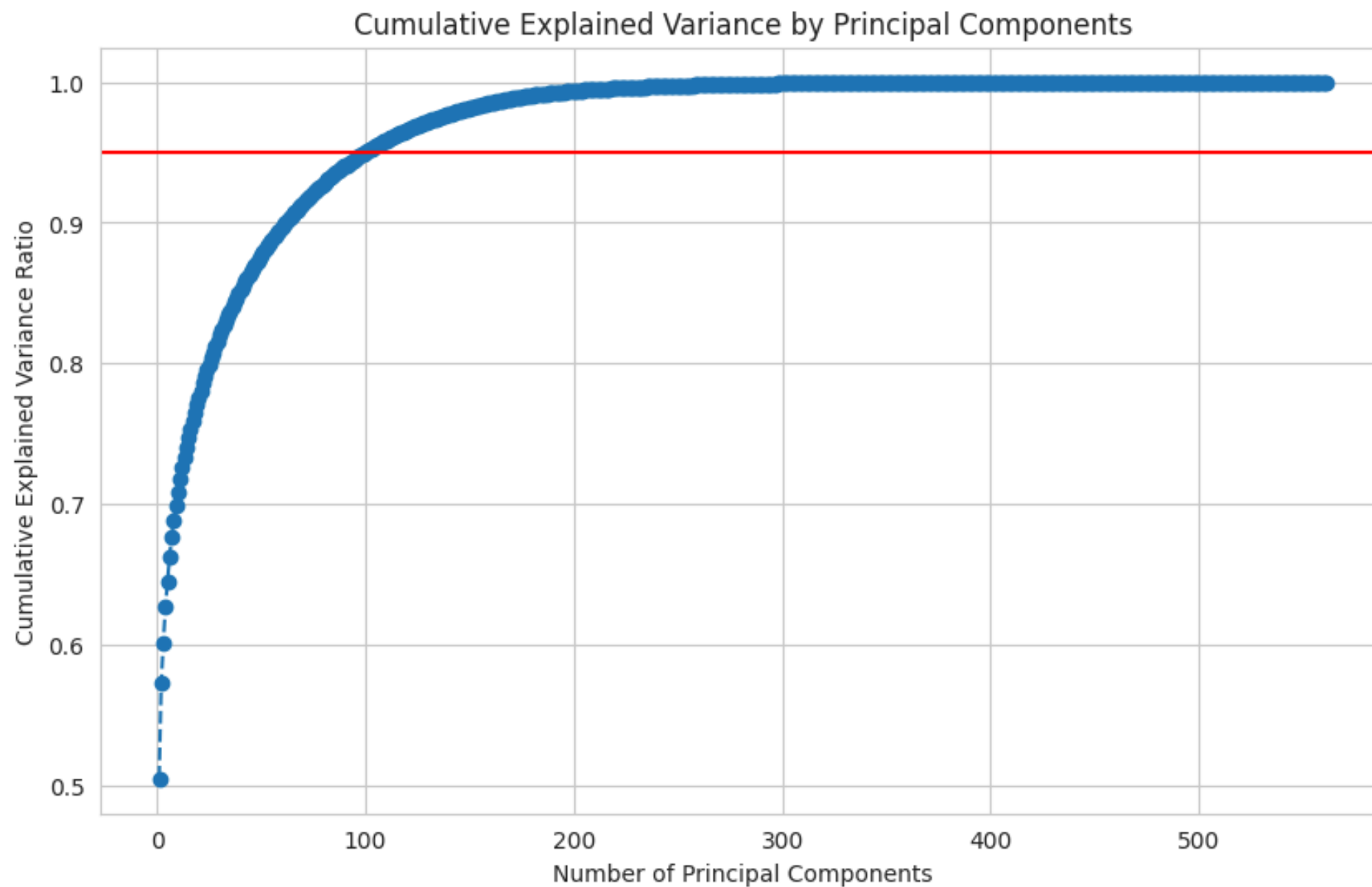


tBodyAcc-mean()-X
tBodyAcc-max()-Z
tBodyAcc-entropy()-X
tBodyAcc-arCoeff()-Z,1
tGravityAcc-std()-Y
tGravityAcc-sma()
tGravityAcc-arCoeff()-X,2
tGravityAcc-correlation()-X,Y
tBodyAccJerk-mad()-Z
tBodyAccJerk-iqr()-X
tBodyAccJerk-arCoeff()-Y,2
tBodyGyro-mean()-Y
tBodyGyro-min()-X
tBodyGyro-entropy()-Y
tBodyGyro-arCoeff()-Z,2
tBodyGyroJerk-std()-Z
tBodyGyroJerk-energy()-X
tBodyGyroJerk-arCoeff()-X,3
tBodyGyroJerk-correlation()-X,Z
tBodyAccMag-arCoeff()1
tGravityAccMag-iqr()
tBodyAccJerkMag-sma()
tBodyGyroMag-max()
tBodyGyroJerkMag-std()
tBodyGyroJerkMag-arCoeff()4
fBodyAcc-max()-Y
fBodyAcc-iqr()-Z
fBodyAcc-kurtosis()-X
fBodyAcc-bandsEnergy()-49,56
fBodyAcc-bandsEnergy()-25,32.1
fBodyAcc-bandsEnergy()-1,8.2
fBodyAcc-bandsEnergy()-49,64.2
fBodyAccJerk-mad()-Z
fBodyAccJerk-iqr()-X
fBodyAccJerk-meanFreq()-Z
fBodyAccJerk-bandsEnergy()-33,40
fBodyAccJerk-bandsEnergy()-9,16.1
fBodyAccJerk-bandsEnergy()-1,24.1
fBodyAccJerk-bandsEnergy()-17,32.2
fBodyGyro-mad()-X
fBodyGyro-energy()-Y
fBodyGyro-meanFreq()-X
fBodyGyro-bandsEnergy()-17,24
fBodyGyro-bandsEnergy()-25,48
fBodyGyro-bandsEnergy()-33,48.1
fBodyGyro-bandsEnergy()-57,64.2
fBodyAccMag-min()
fBodyBodyAccJerkMag-mad()
fBodyBodyGyroMag-mean()
fBodyBodyGyroMag-skewness()
fBodyBodyGyroJerkMag-maxInds

Correlation Matrix of the Features



FEATURE ELIMINATION



561 FEATURE

PCA

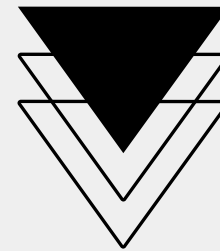
VARIANCE > %95

100 FEATURE

CLASSIFICATION METHODS

*Simple Neural
Network*

CNN



SVM

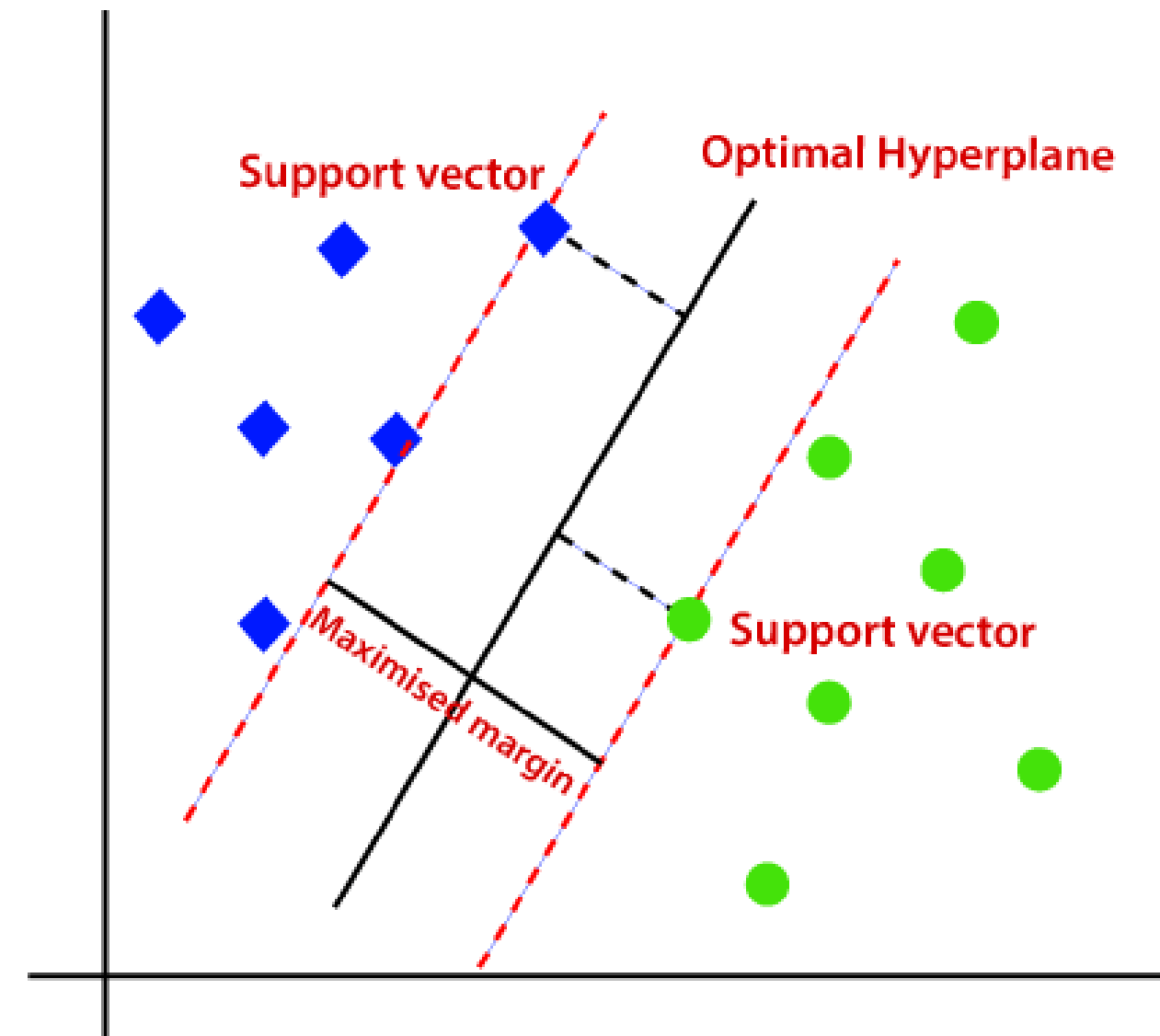
SVM

KERNEL "RBF"

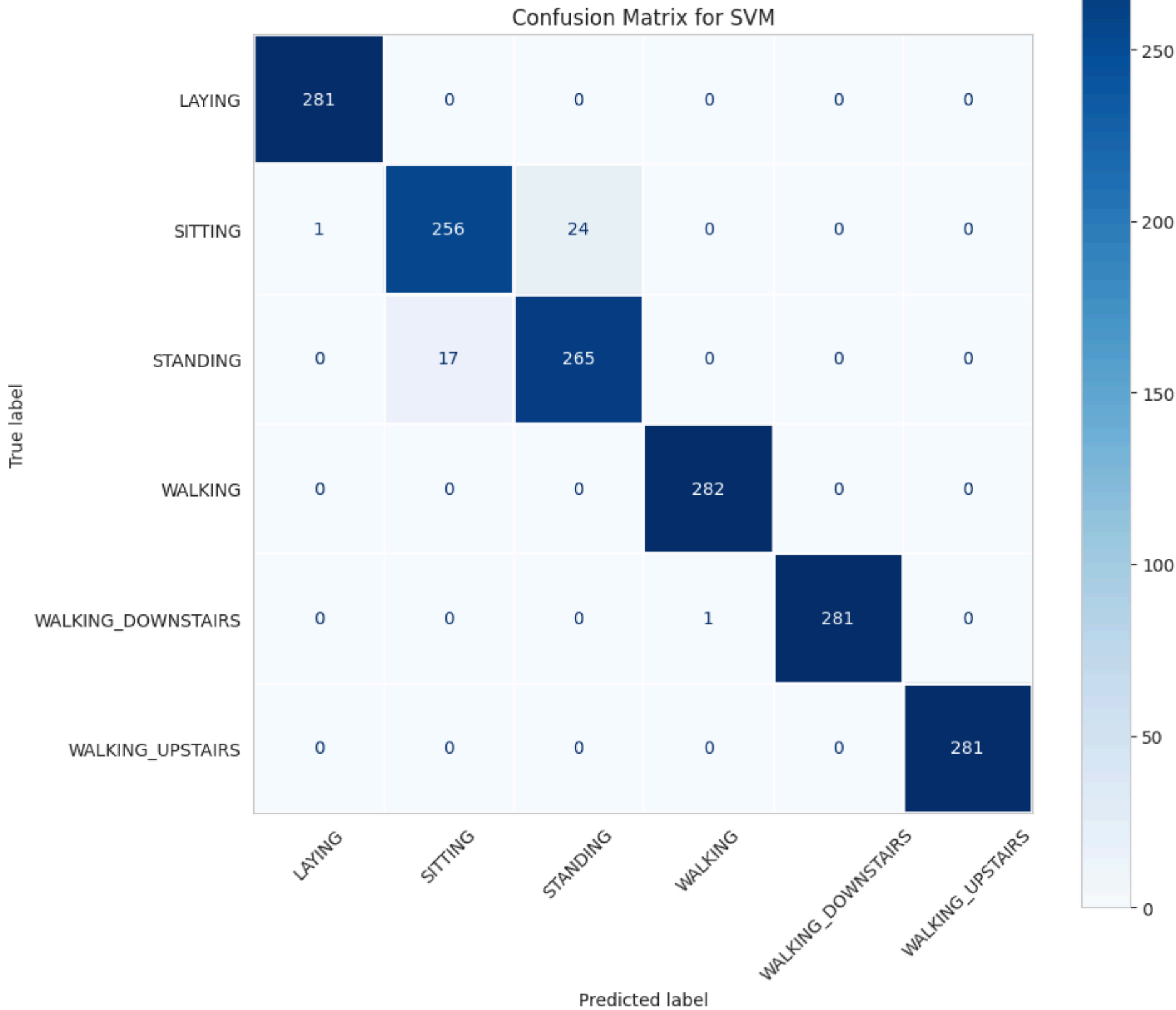
$C=1$

ACCURACY = 0.9745

F1 SCORE = 0.9745

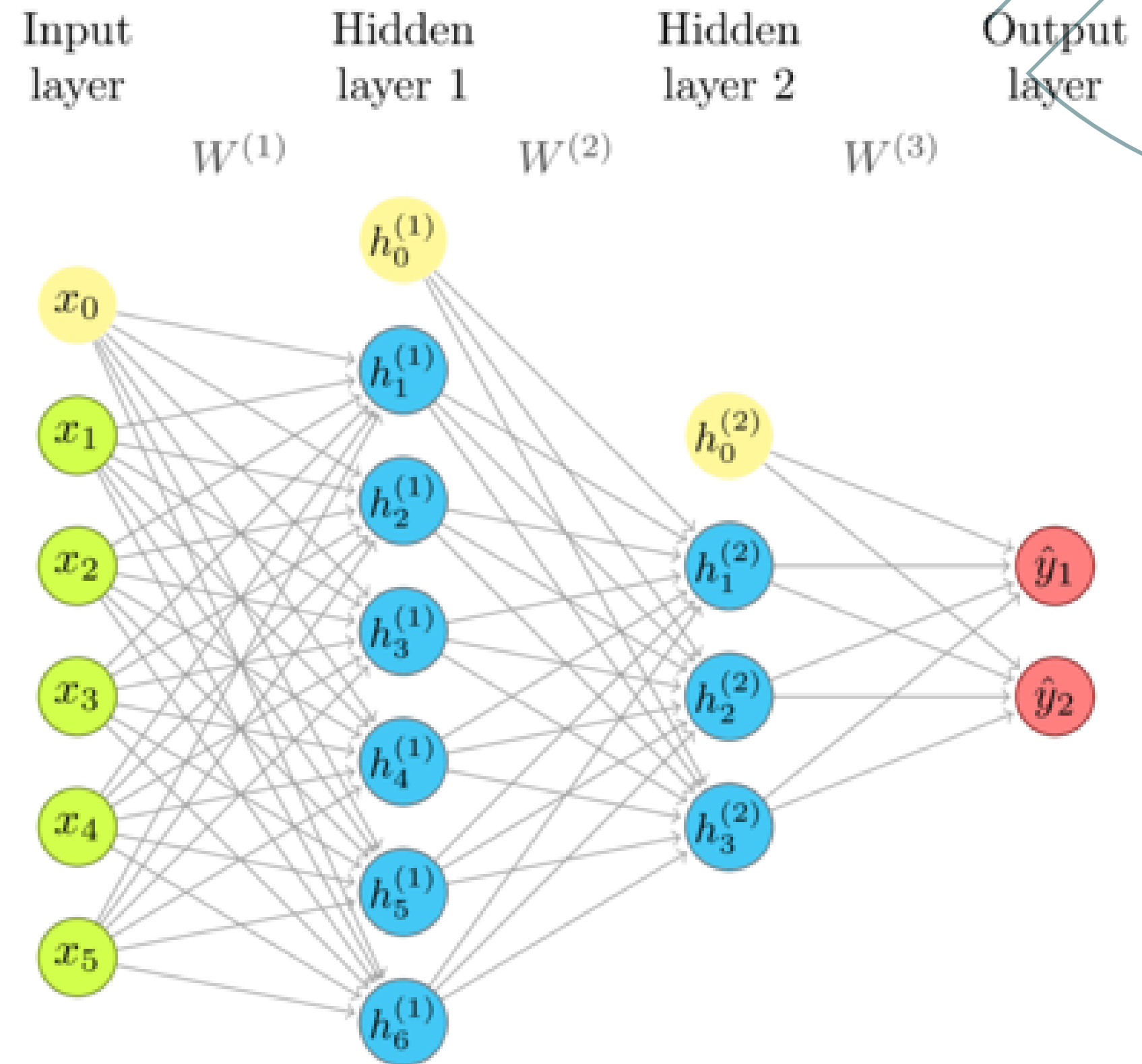


SVM RESULTS



SIMPLE NN

- 3 fully connected
- Dropout layer (probability of 0.5)
- **Relu** activation function
- 100 epochs
- **Cross-Entropy Loss**
- **Adam optimizer**
- **5 fold Cross Validation**



SIMPLE NN RESULTS

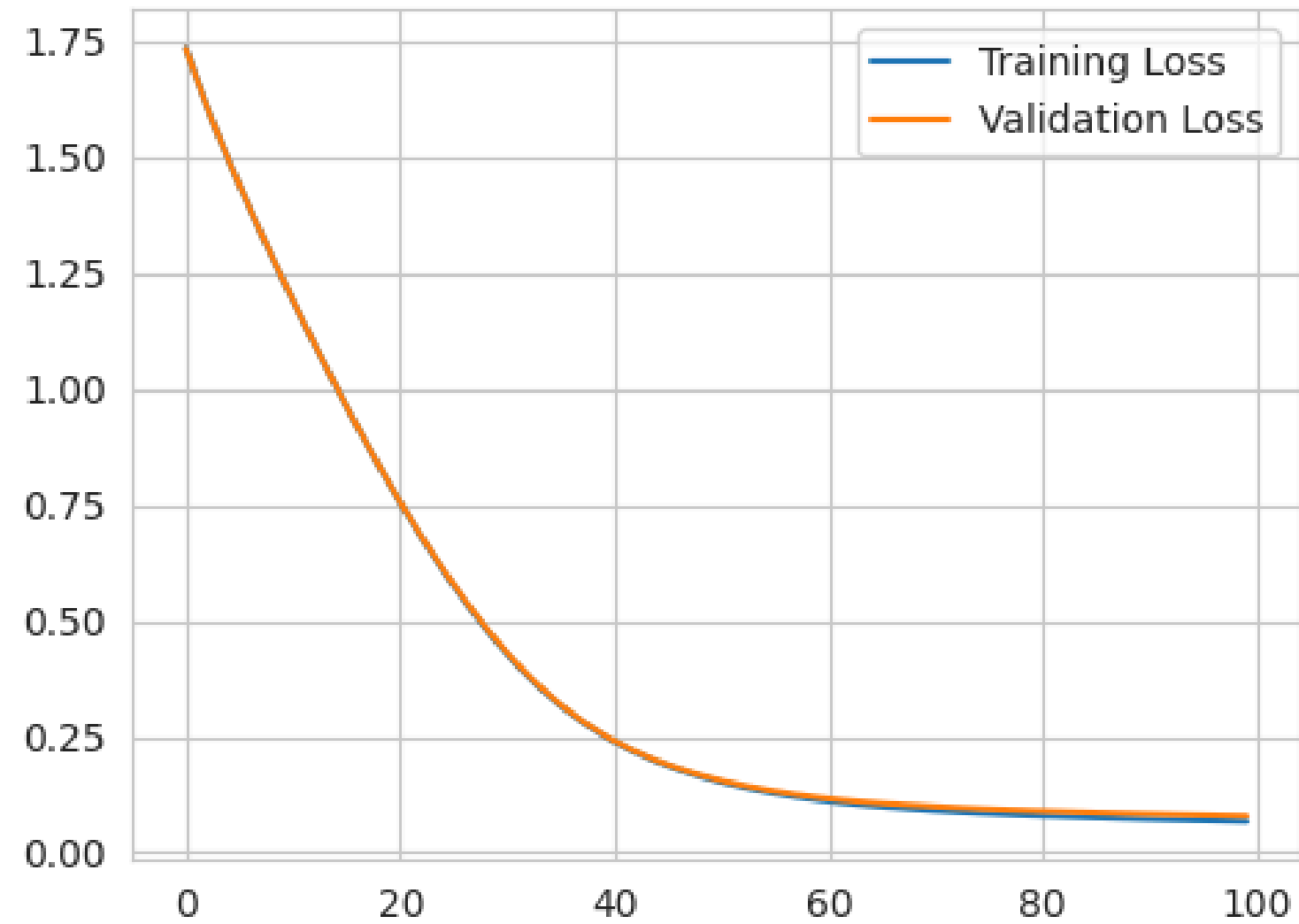
	TRAIN ACCURACY	VALIDATION ACCURACY	PRECISION	F1 SCORE
FOLD 1	0.9757	0.9639	0.96	0.96
FOLD 2	0.9748	0.9745	0.97	0.97
FOLD 3	0.9720	0.9627	0.96	0.96
FOLD 4	0.9742	0.9621	0.96	0.96
FOLD 5	0.9754	0.9698	0.97	0.97
AVERAGE	0.9744	0.9666	0.9744	0.9665

ACCURACY = 0.9745

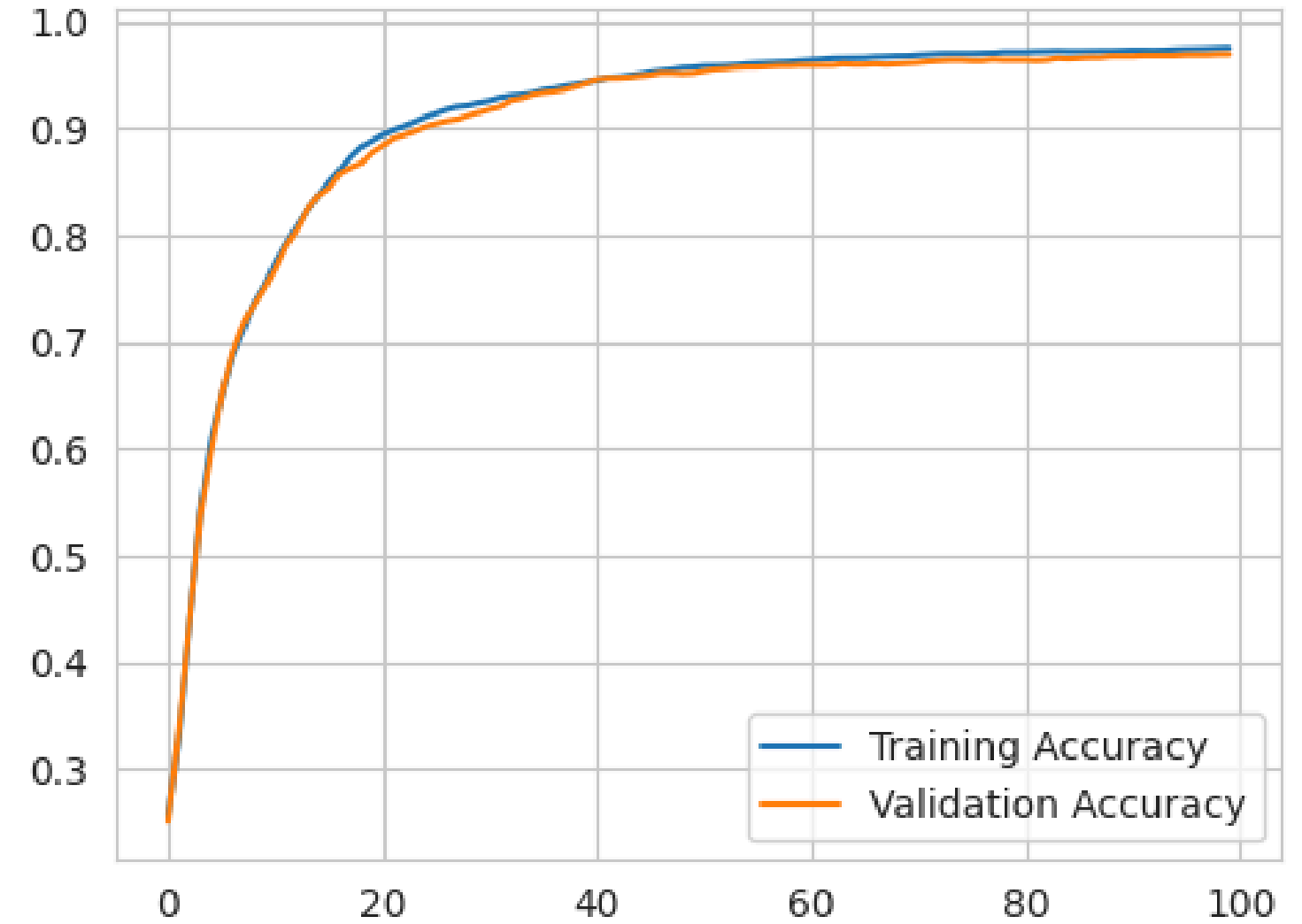
F1 SCORE = 0.9665

SIMPLE NN RESULTS

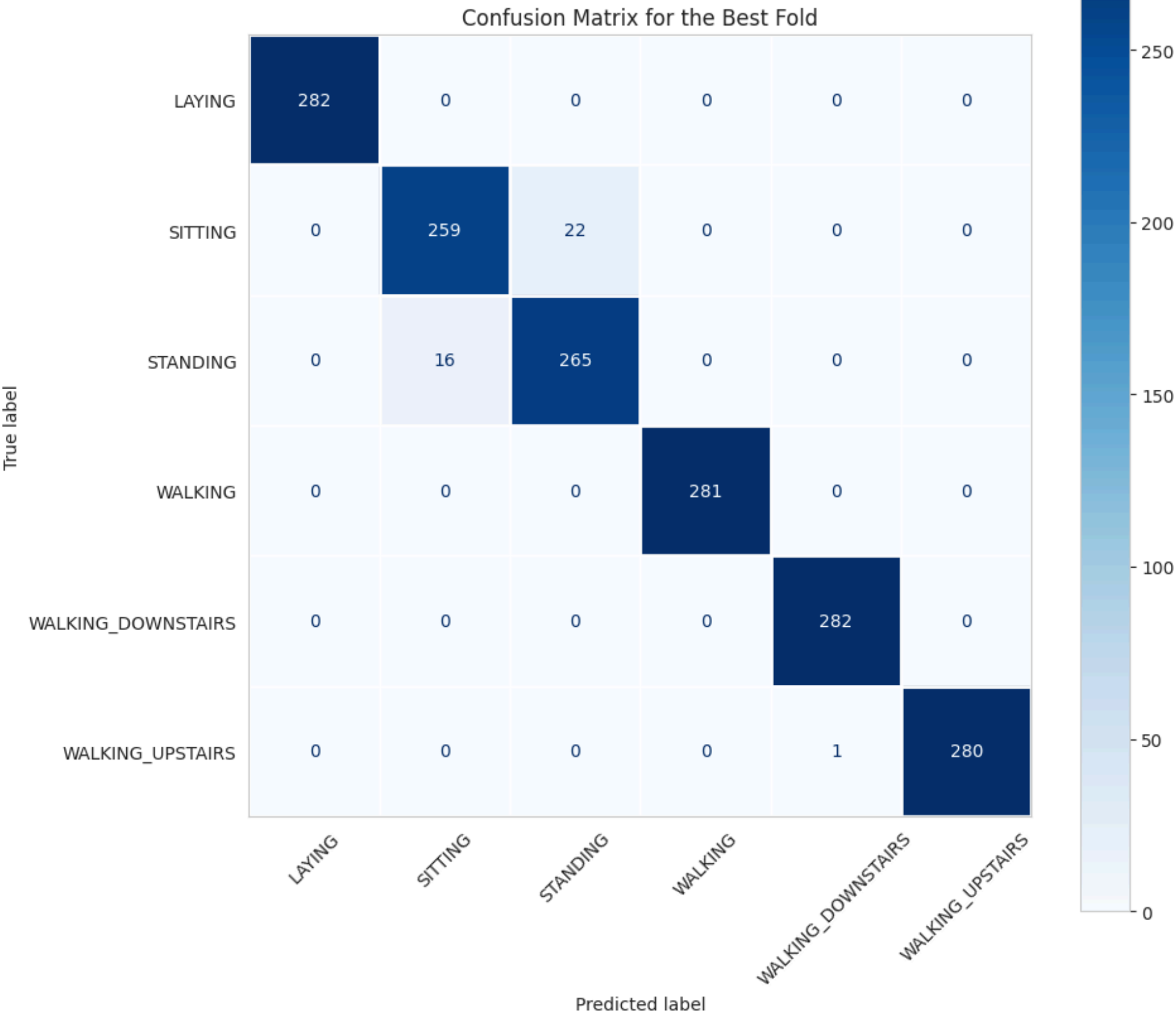
Training and Validation Loss



Training and Validation Accuracy

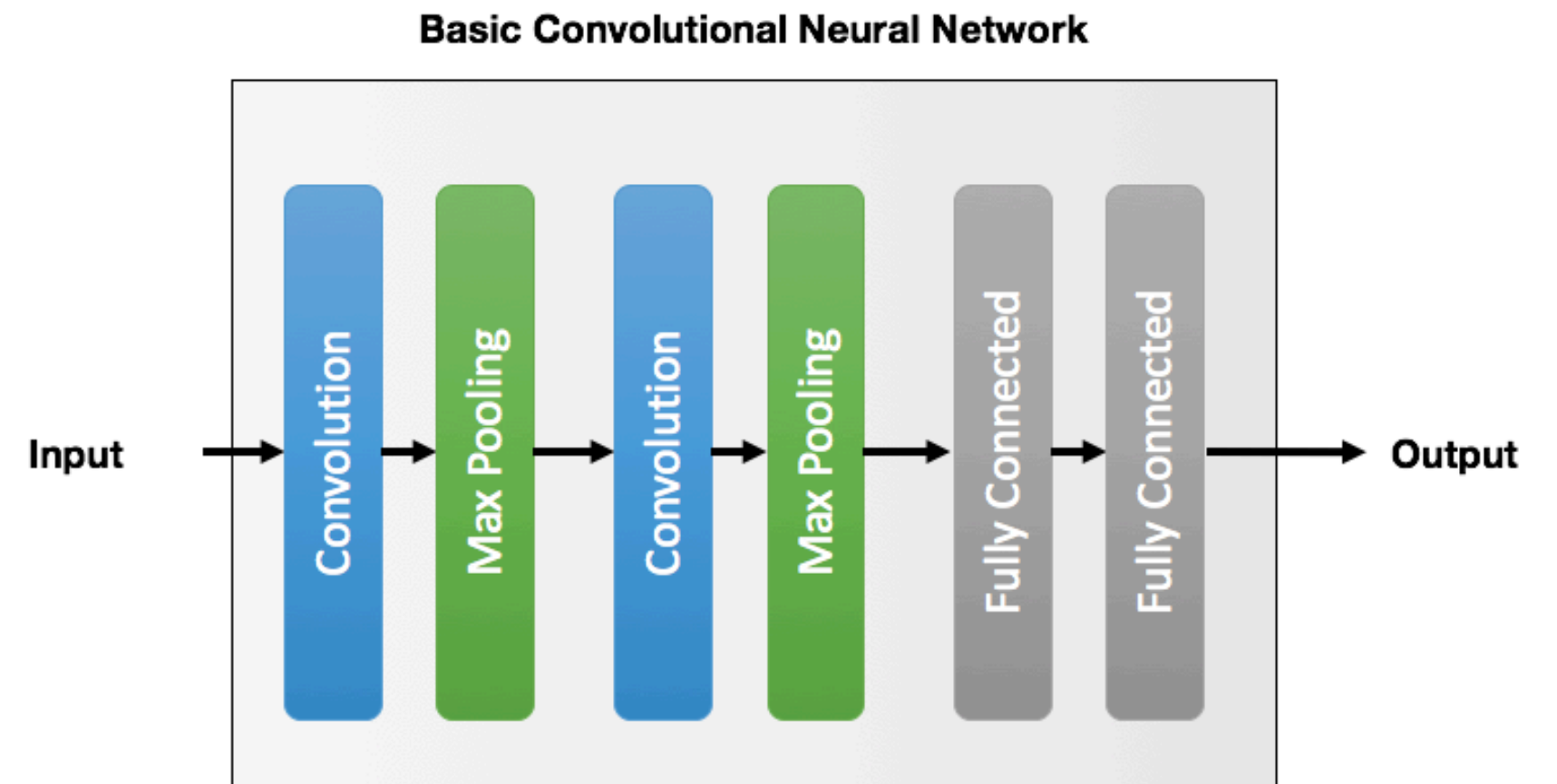


SIMPLE NN RESULTS



TRADITIONAL CNN

- 2 Convolution layer
 - Relu activation function
 - Max pooling
 - Dropout layer (probability of 0.5)
 - 2 Fully connected layer
 - Flattening Layer
- 100 epochs
 - **Cross-Entropy Loss**
 - **Adam optimizer**
 - **5 fold Cross Validation**



TRADITIONAL CNN

Layer Type	Layer Name	Parameters
Input Layer	-	-
Convolutional	conv1	in_channels=1, out_channels=32, kernel_size=3, stride=1, padding=1
Activation	relu	-
Max Pooling	maxpool	kernel_size=2, stride=2
Convolutional	conv2	in_channels=32, out_channels=64, kernel_size=3, stride=1, padding=1
Activation	relu	-
Max Pooling	maxpool	kernel_size=2, stride=2
Flattening	-	-
Fully Connected	fc1	in_features=flattened_size, out_features=128
Activation	relu	-
Dropout	dropout	p=0.5
Fully Connected	fc2	in_features=128, out_features=num_classes

CNN RESULTS

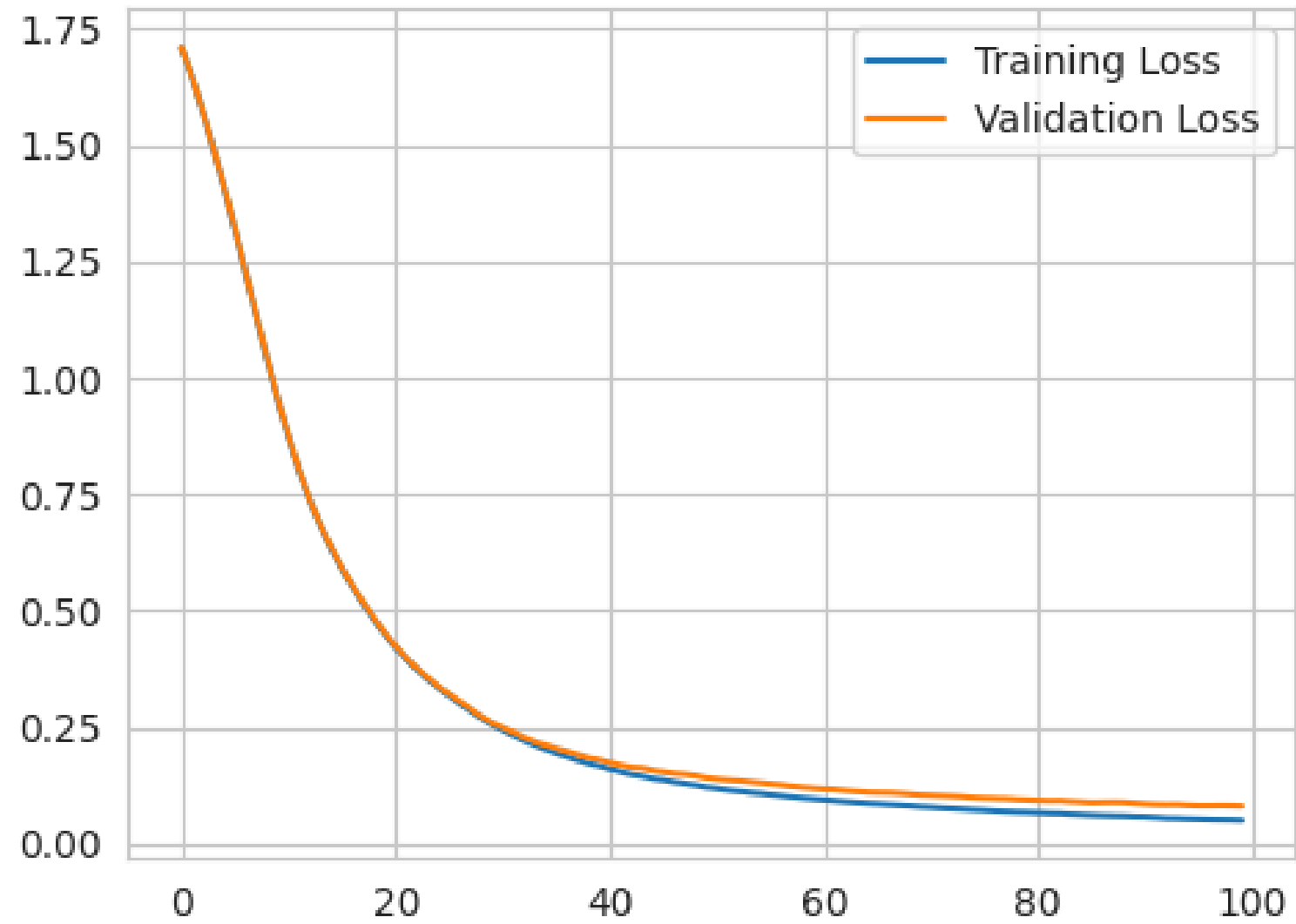
	TRAIN ACCURACY	VALIDATION ACCURACY	PRECISION	F1 SCORE
FOLD 1	0.9815	0.9615	0.96	0.96
FOLD 2	0.9825	0.9757	0.98	0.98
FOLD 3	0.9837	0.9609	0.96	0.96
FOLD 4	0.9822	0.9591	0.96	0.96
FOLD 5	0.9842	0.9668	0.97	0.97
AVERAGE	0.9828	0.9648	0.9744	0.9648

ACCURACY = 0.9828

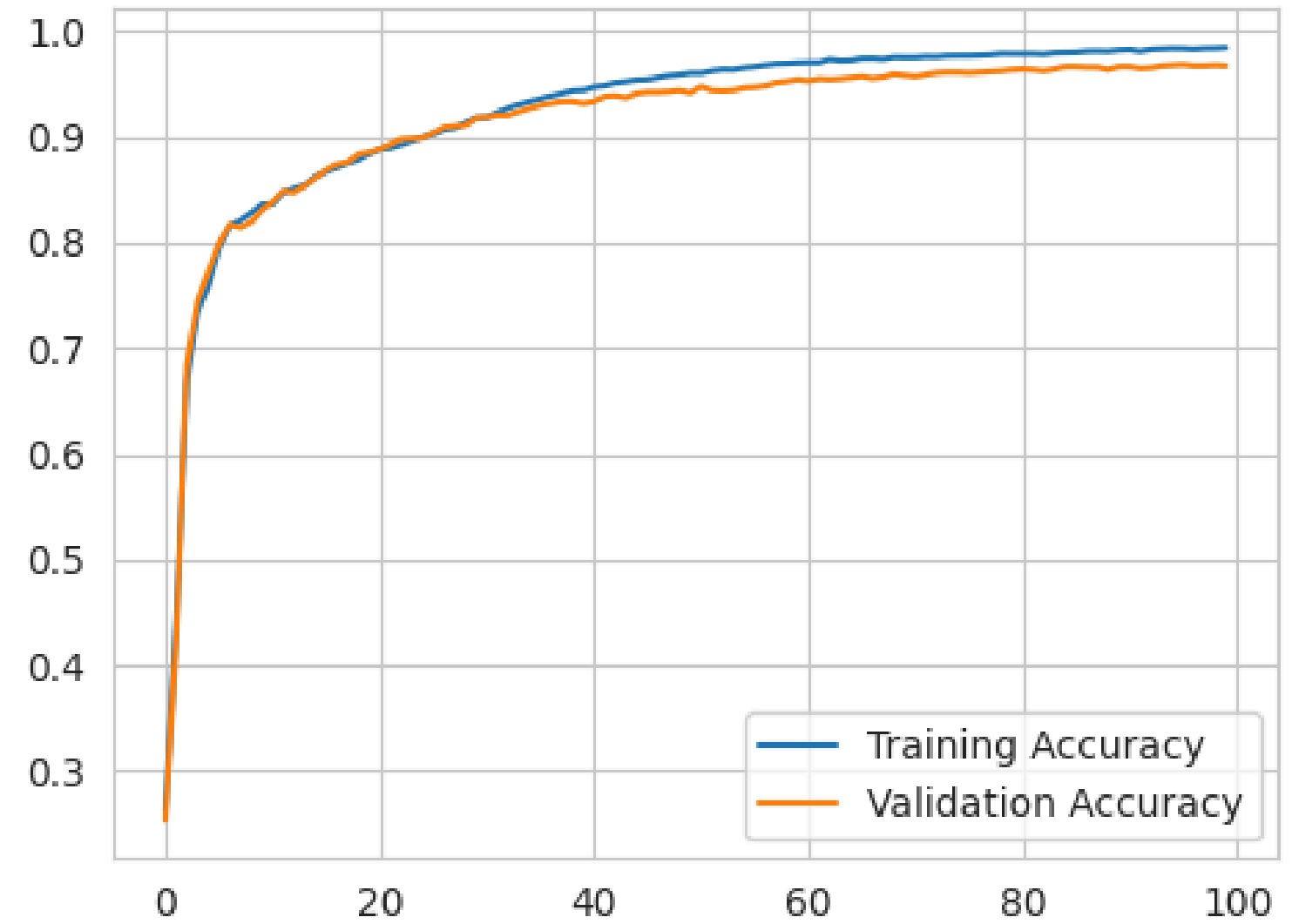
F1 SCORE = 0.9648

CNN RESULTS

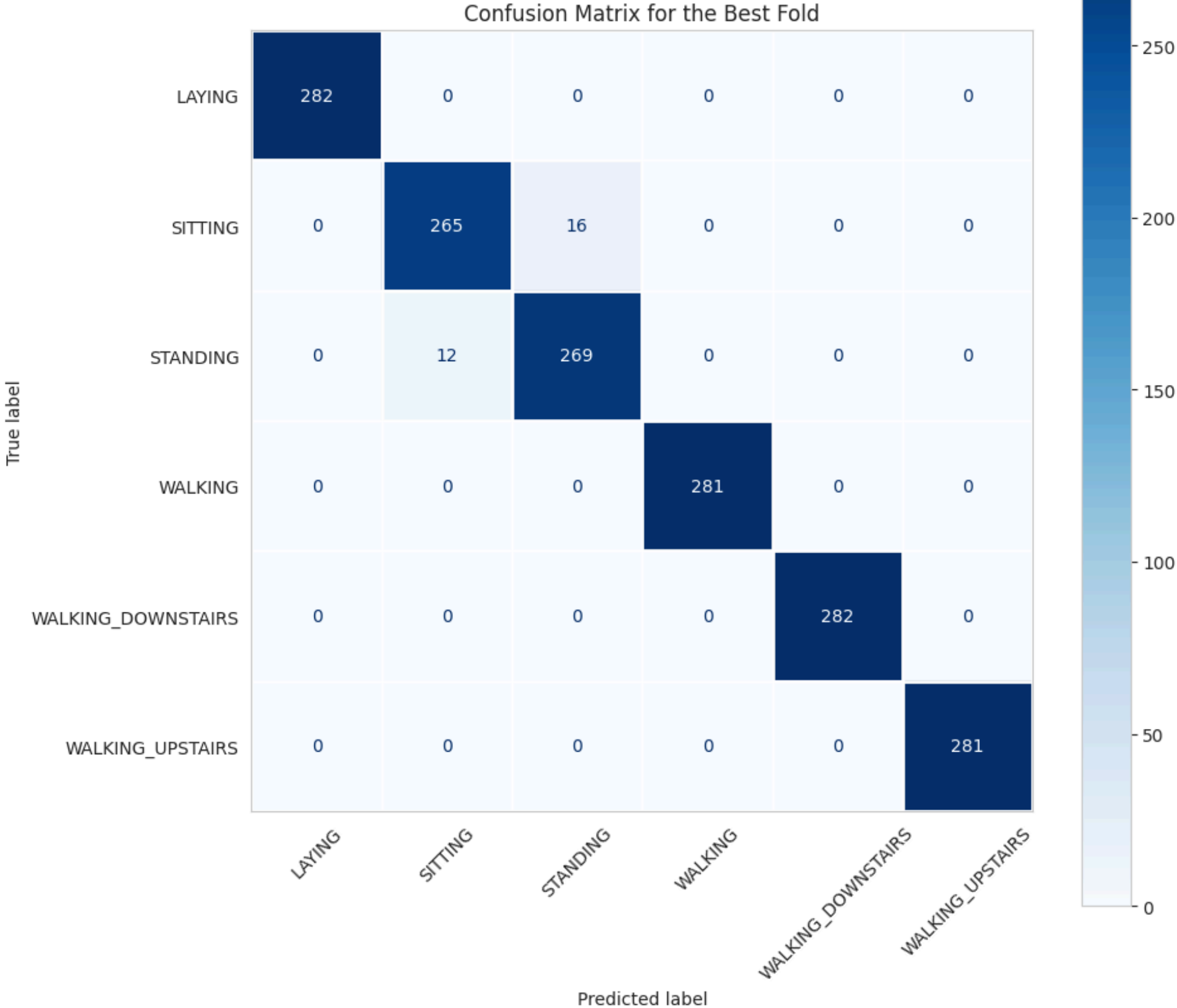
Training and Validation Loss



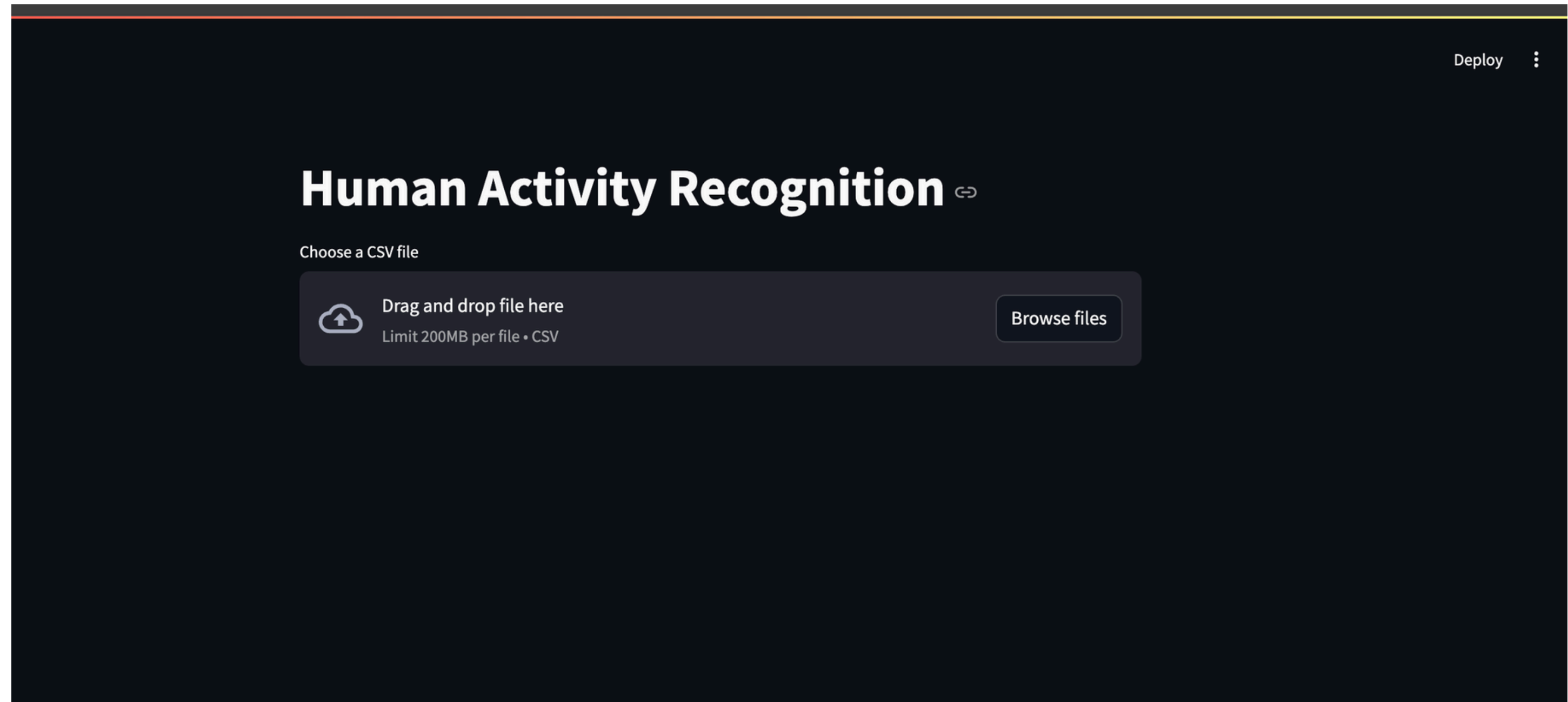
Training and Validation Accuracy



CNN RESULTS



DASHBOARD



Human Activity Recognition

Choose a CSV file



Drag and drop file here

Limit 200MB per file • CSV

Browse files



test.csv 19.3MB

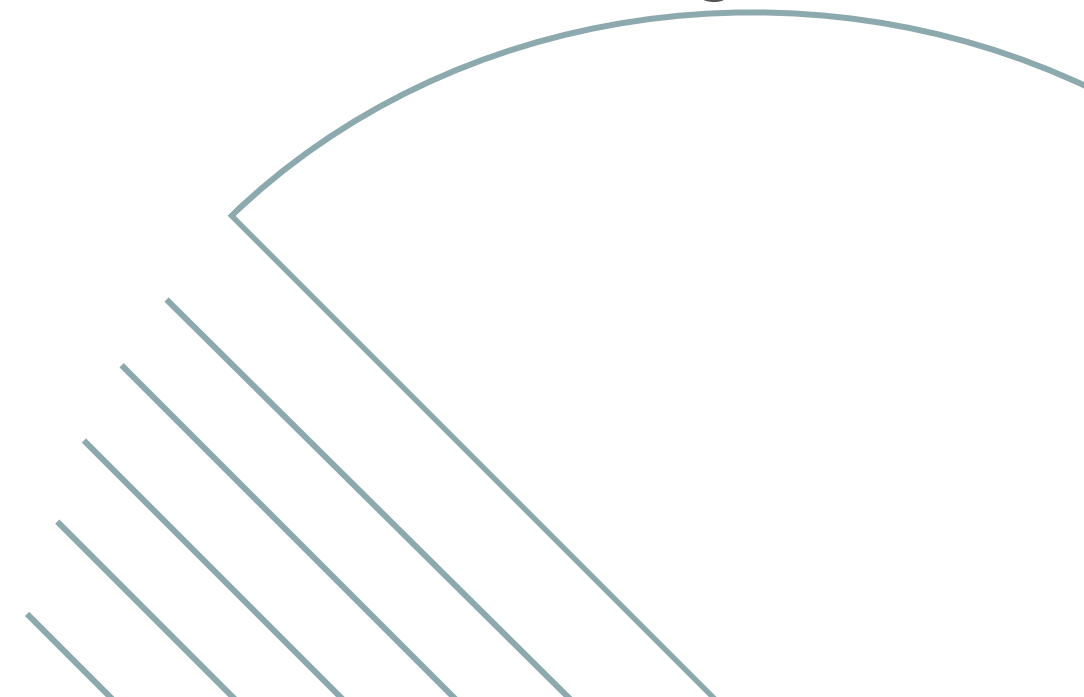
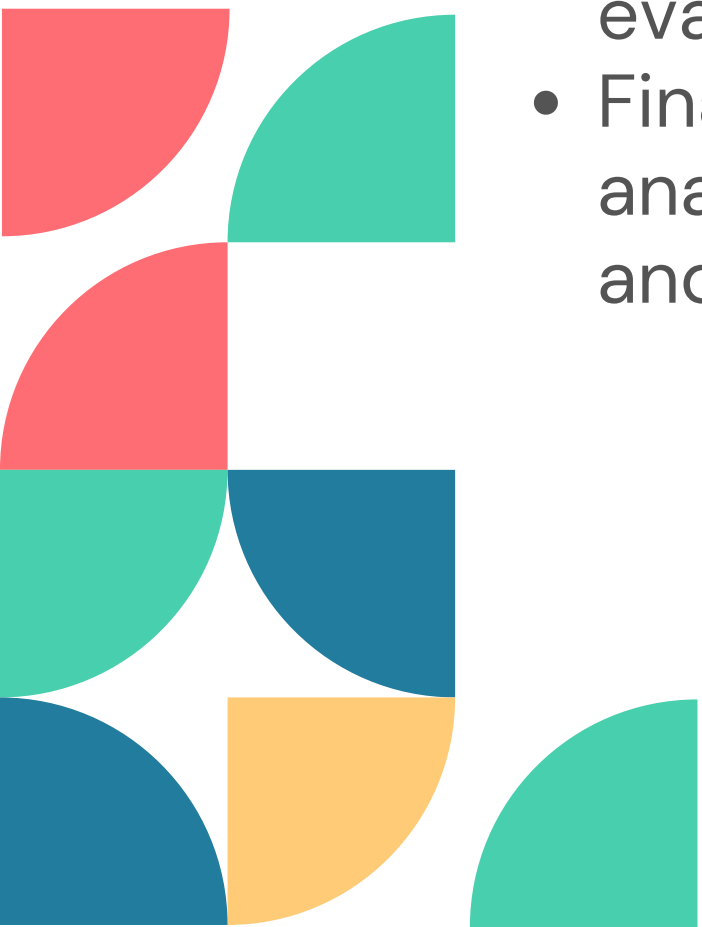


Classification Report:

	precision	recall	f1-score	support
LAYING	0.6029	0.3054	0.4054	537
SITTING	0.3808	0.6701	0.4856	491
STANDING	0.442	0.3365	0.3821	532
WALKING	0.5492	0.506	0.5268	496
WALKING_DOWNSTAIRS	0.3814	0.4405	0.4088	420
WALKING_UPSTAIRS	0.6832	0.673	0.6781	471
accuracy	0.4835	0.4835	0.4835	0.4835
macro avg	0.5066	0.4886	0.4811	2,947
weighted avg	0.5091	0.4835	0.4791	2,947

SUMMARY

- The project aims to enhance human activity recognition using sensor data through various machine learning techniques, including SVM, Simple NN, and CNN, integrated into an interactive dashboard.
- Initially, data preprocessing involves with standardizing features. PCA is then applied to reduce dimensionality, preserving essential variance.
- An SVM model with an RBF kernel is trained and evaluated using accuracy, F1 score, and confusion matrix,
- A Simple NN and a CNN are also developed, each with distinct architectures involving layers for input, hidden processing, activation, and dropout, followed by evaluation and visualization.
- Finally, an interactive Streamlit dashboard is prepared, enabling real-time data analysis and model prediction, providing a comprehensive tool for understanding and deploying activity recognition models



The image features a light gray background with the text "THANK YOU" centered in a bold, blue, sans-serif font. The corners are decorated with abstract geometric patterns. The top-left corner has a series of parallel diagonal lines in a light blue-gray color. The top-right corner features a cluster of overlapping semi-circles in yellow, red, teal, and dark blue. The bottom-left corner also has a cluster of overlapping semi-circles in red, teal, and dark blue. The bottom-right corner contains a large, light blue-gray arc with several parallel diagonal lines extending from its base.

THANK YOU